# ETALE COVER BY PIECES THAT ARE FLAT WITH GEOMETRICALLY CONNECTED AND REDUCED FIBRES

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 13269-13324, immediately following the statement at 13257-13267.

*Editorial note.* Only two cross-chapter pointers (Varieties) are needed and they are recorded in assembly.external_references.

*Editorial note.* The proof reduces to a case handled 'in the next paragraph' inside the same span; no material is left outside the counted proof.

*Allowed standard prerequisites (editorial).*

- Fibres geometrically connected and reduced after shrinking. chapter lemma lemma-fibre-geometrically-connected-reduced, cited at proof line 13318. Produces the affine etale neighbourhood S' and the quasi-compact open X' with geometrically connected fibres over S'.
- Realising prescribed residue field extensions by etale covers. chapter lemma lemma-realize-prescribed-residue-field-extension-etale, cited in the proof. Lets the field extension needed to separate the connected components of the fibre be realised etale over the base.
- Characterisation of geometric connectedness of a fibre. Varieties, lemma varieties-lemma-characterize-geometrically-connected, cited in the proof. Turns the condition on a fibre into an algebraic statement about the residue field extension, which is what the etale cover is built to satisfy.
- Galois action on a quasi-compact open. Varieties, lemma varieties-lemma-Galois-action-quasi-compact-open, cited in the proof. Controls the open pieces of the fibre under the finite group action arising from the chosen finite extension.

*Source reference.* Original lemma 055W: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13193–13193. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 02LF: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10138–10138. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 04KU: focused-seeds/geometry/S9-morph-cover-by-geom-connected/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Date*: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original reference 0387: focused-seeds/geometry/S9-morph-cover-by-geom-connected/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $f : X \to S$ be a morphism of schemes which is locally of finite presentation and flat with geometrically reduced fibres. Then there exists an étale covering $\{X_i \to X\}_{i \in I}$ such that $X_i \to S$ factors as $X_i \to S_i \to S$ where $S_i \to S$ is étale and $X_i \to S_i$ is flat of finite presentation with geometrically connected and geometrically reduced fibres.*

*Proof.* Pick a point $x \in X$ with image $s \in S$. We will produce a diagram

$$\begin{array}{ccccc} X' & \longrightarrow & S' \times_S X & \longrightarrow & X \\ & \searrow & \downarrow & & \downarrow \\ & & S' & \longrightarrow & S \end{array}$$

and points $s' \in S'$, $x' \in X'$, $y \in S' \times_S X$ such that $x'$ maps to $x$, $(S', s') \to (S, s)$ is an étale neighbourhood, $(X', x') \to (S' \times_S X, y)$ is an étale neighbourhood[1], and $X' \to S'$ has geometrically connected fibres. If we can do this for every $x \in X$, then the lemma follows (with members of the covering given by the collection of étale morphisms $X' \to X$ so produced). The first step is the replace $X$ and $S$ by affine open neighbourhoods of $x$ and $s$ which reduces us to the case that $X$ and $S$ are affine (and hence $f$ of finite presentation).

Choose a separable algebraic extension $\overline{k}$ of $\kappa(s)$. Denote $X_{\overline{k}}$ the base change of $X_s$. Choose a point $\overline{x}$ in $X_{\overline{k}}$ mapping to $x \in X_s$. Choose a connected quasi-compact open neighbourhood $\overline{V} \subset X_{\overline{k}}$ of $\overline{x}$. (This is possible because any scheme locally of finite type over a field is locally connected as a locally Noetherian topological space.) By Varieties, Lemma 04KU we can find a finite separable extension $k'/\kappa(s)$ and a quasi-compact open $V' \subset X_{k'}$ whose base change is $\overline{V}$. In particular $V'$ is geometrically connected over $k'$, see Varieties, Lemma 0387. By Lemma 02LF we can find an étale neighbourhood $(S', s') \to (S, s)$ such that $\kappa(s')$ is isomorphic to $k'$ as an extension of $\kappa(s)$. Denote $x' \in (S' \times_S X)_{s'} = X_{k'}$ the image of $\overline{x}$. Thus after replacing $(S, s)$ by $(S', s')$ and $(X, x)$ by $(S' \times_S X, x')$ we reduce to the case handled in the next paragraph.

Assume there is a quasi-compact open $V \subset X_s$ which contains $x$ and is geometrically connected. Then we can apply Lemma 055W to find an affine étale neighbourhood $(S', s') \to (S, s)$ and a quasi-compact open $X' \subset S' \times_S X$ such that $X' \to S'$ has geometrically connected fibres and such that $X'$ contains a point mapping to $x$. This finishes the proof. □

[1]The proof actually gives an open $X' \subset S' \times_S X$.